Ethanol Blending in Gasoline – Bulgaria

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U.S. GRAINS & BIOPRODUCTS COUNCIL

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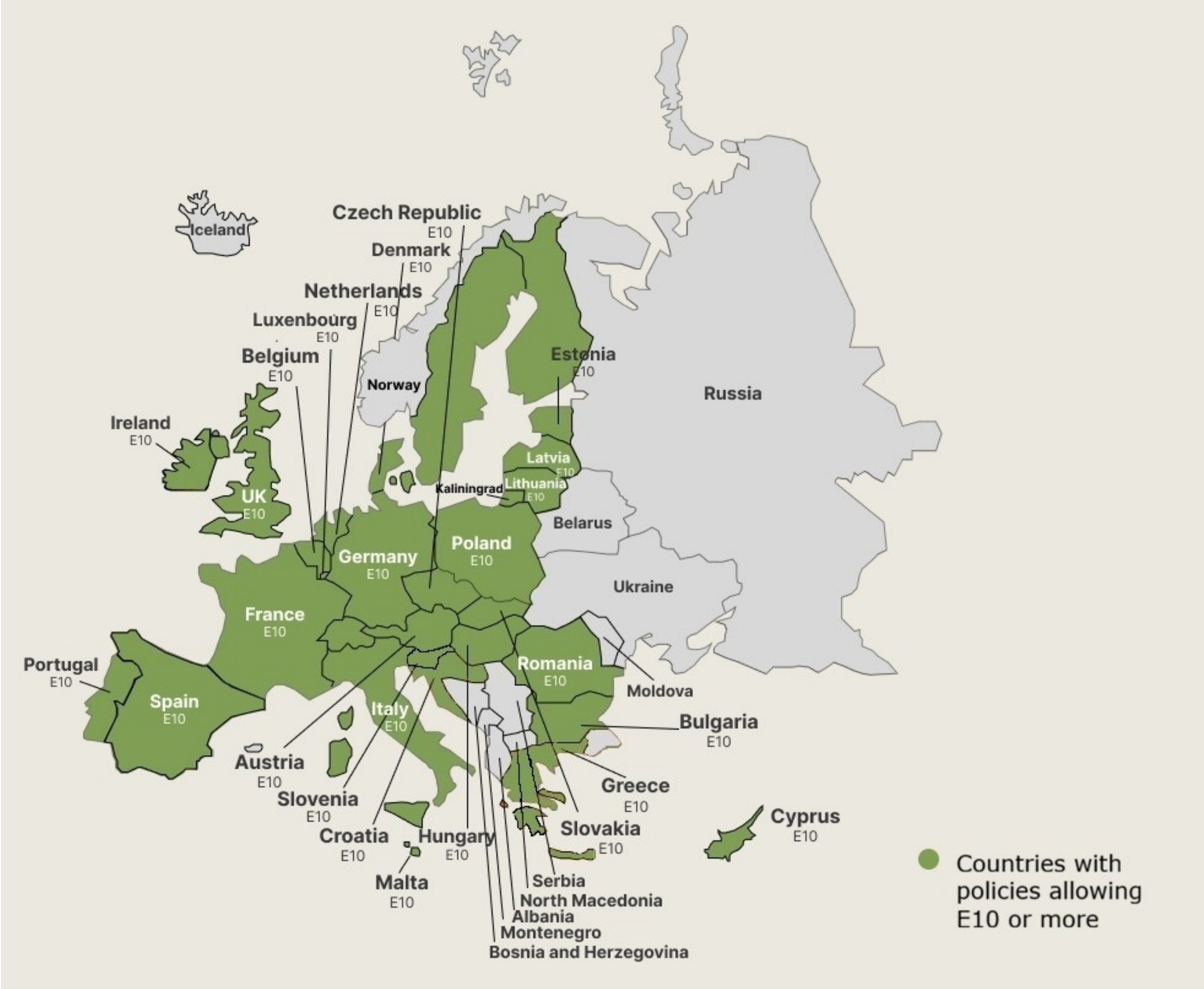
Ethanol Blending in Europe

There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

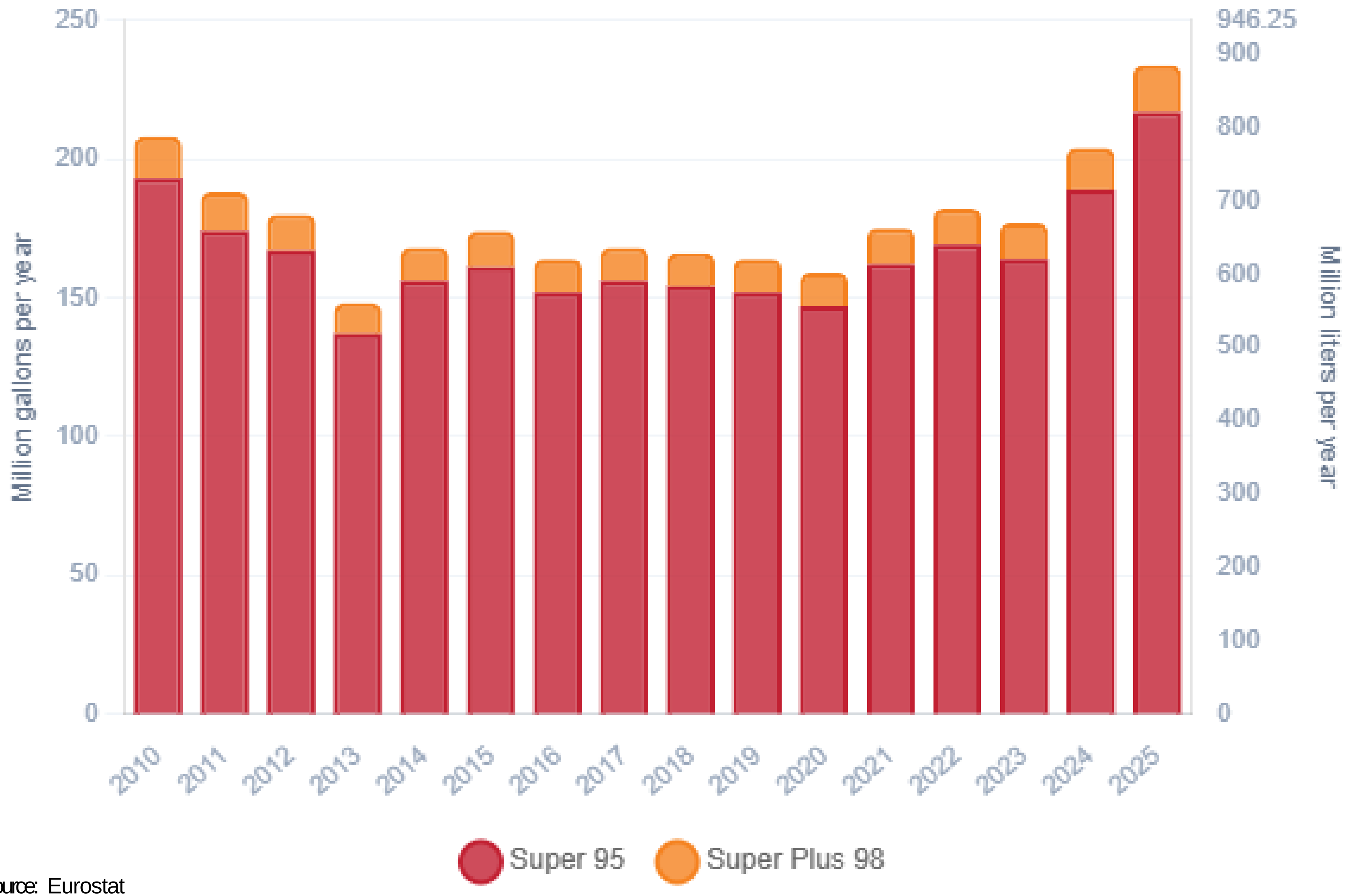
Ethanol Gasoline Blending in Bulgaria

In 2025, gasoline consumption in Bulgaria was 885 million liters (234 million gallons) and growth rate was 0.79%. Gasoline production and net imports were 1,664 and -735 million liters (440 and -194 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 6% of gasoline demand. Ethanol Net Imports were 15 million liters each (4 million gallons). Ethanol sources are Corn 45% and Cereals 55%.

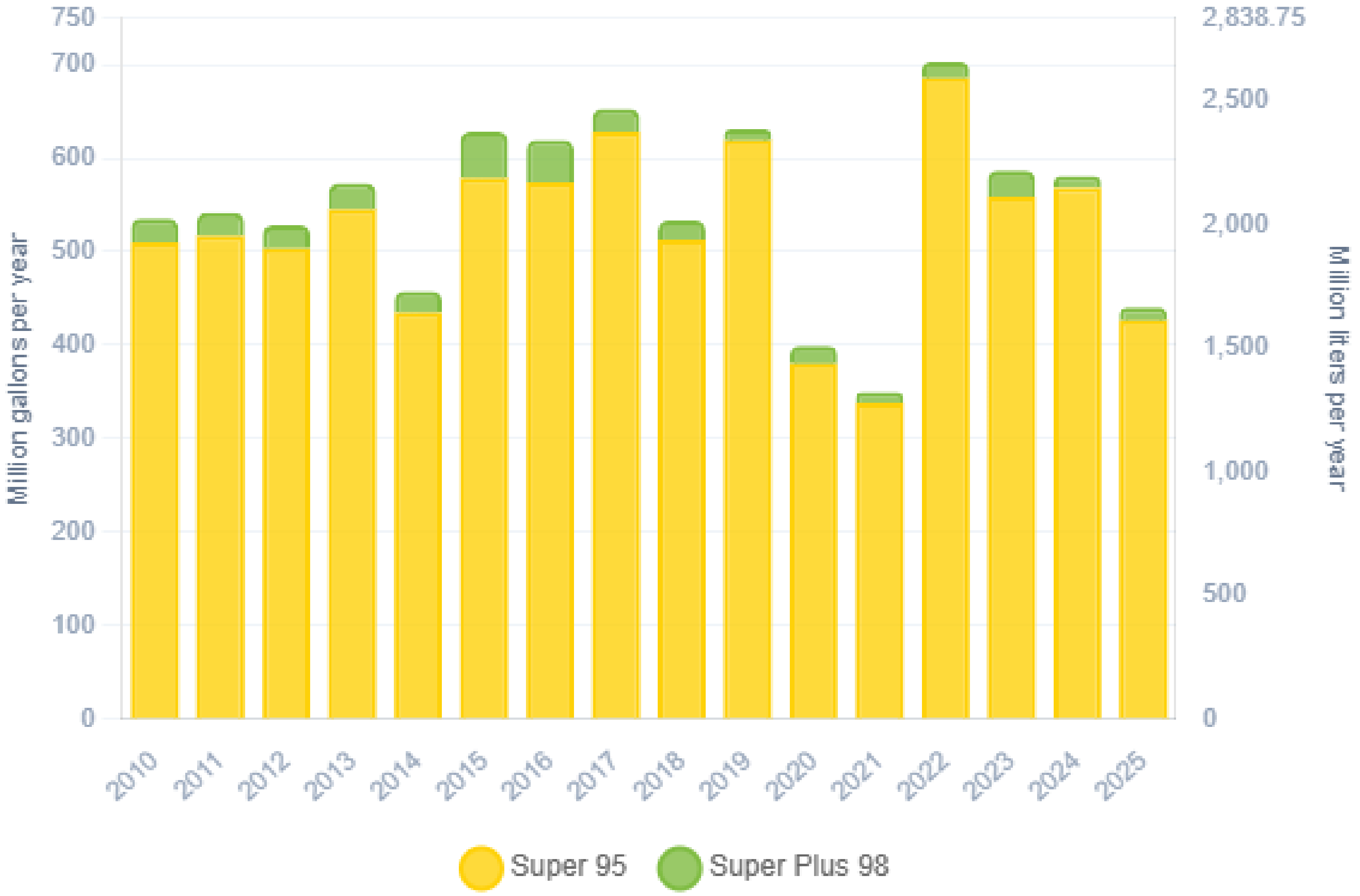
Source: Eurostat, European Commision

Gasoline Demand in Bulgaria



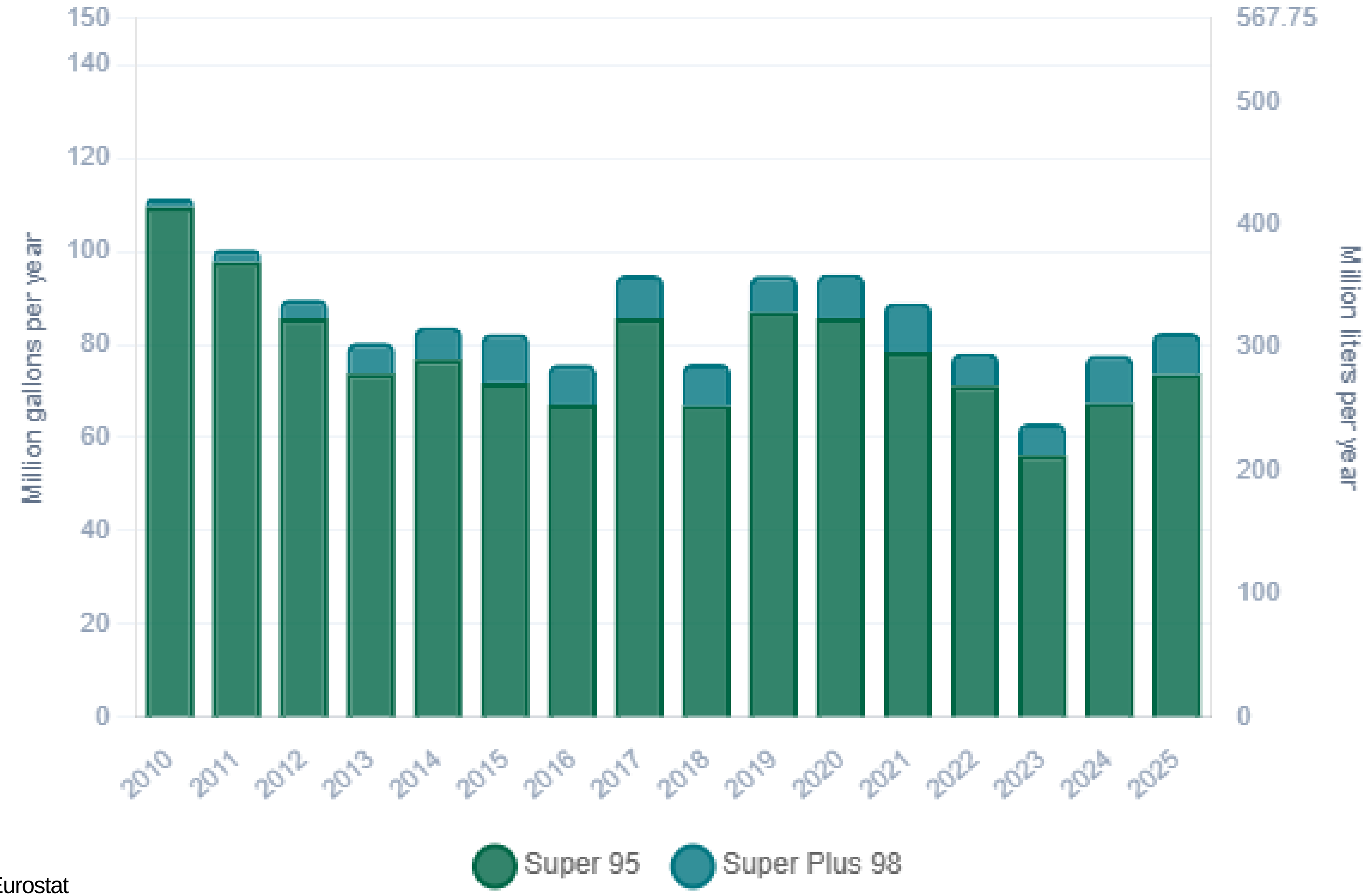
Source: Eurostat

Gasoline Production in Bulgaria



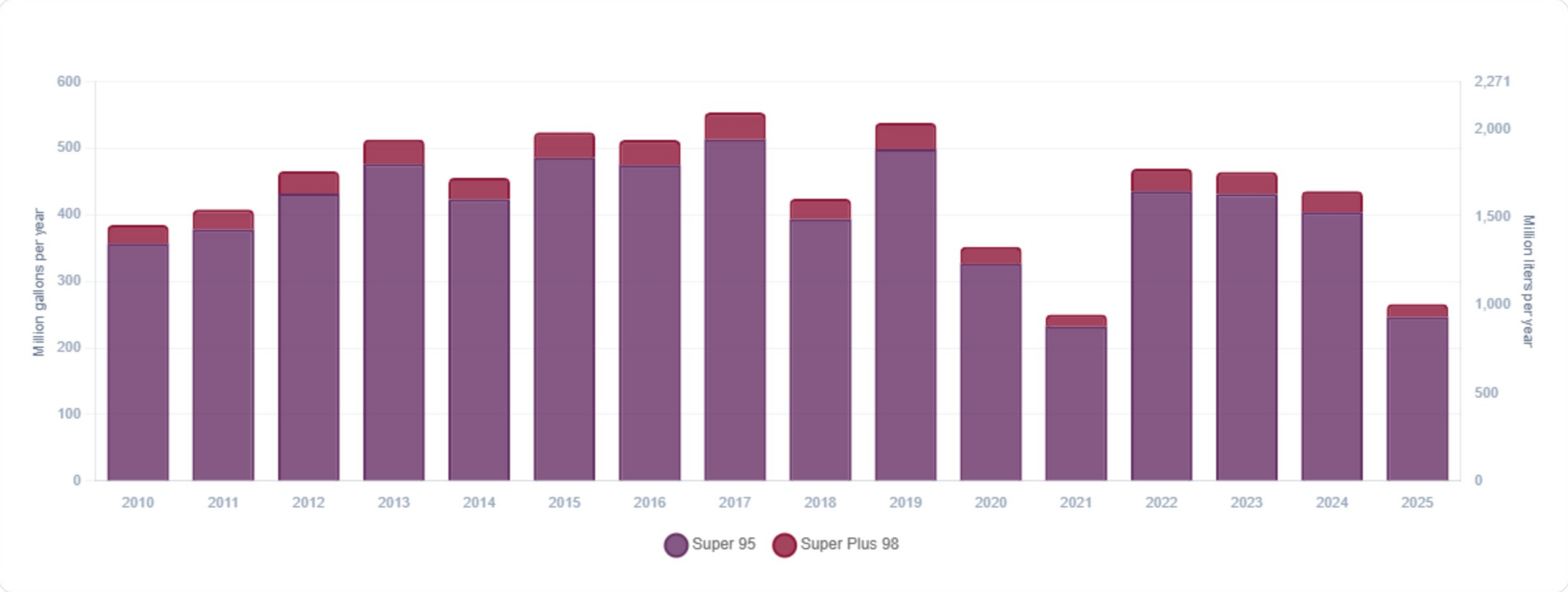
Source: Eurostat

Gasoline Imports in Bulgaria



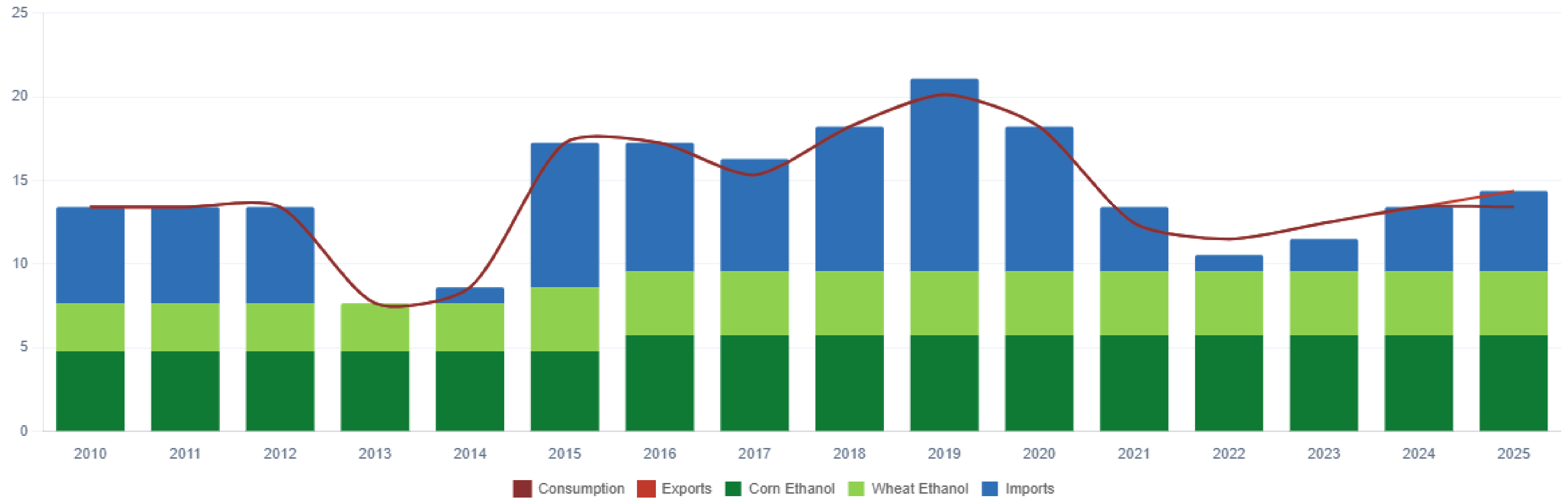
Source: Eurostat

Gasoline Exports in Bulgaria



Source: Eurostat

Ethanol Balance in Bulgaria



Source: Eurostat

Gasoline Quality in Bulgaria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

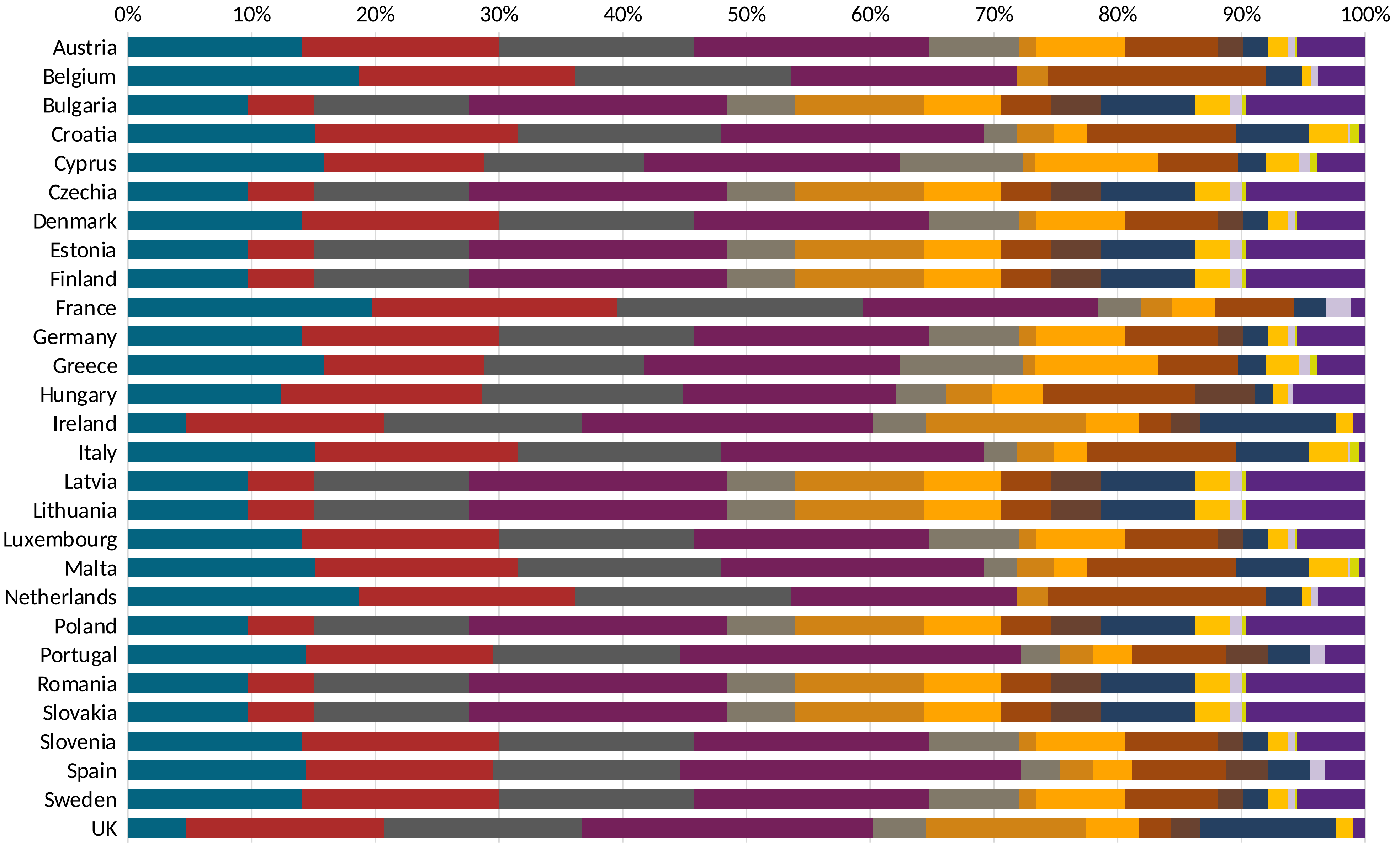
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Gasoline Component Blending in Europe

Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naptha
- Heavy Primary Naptha
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics



Source: Faro90

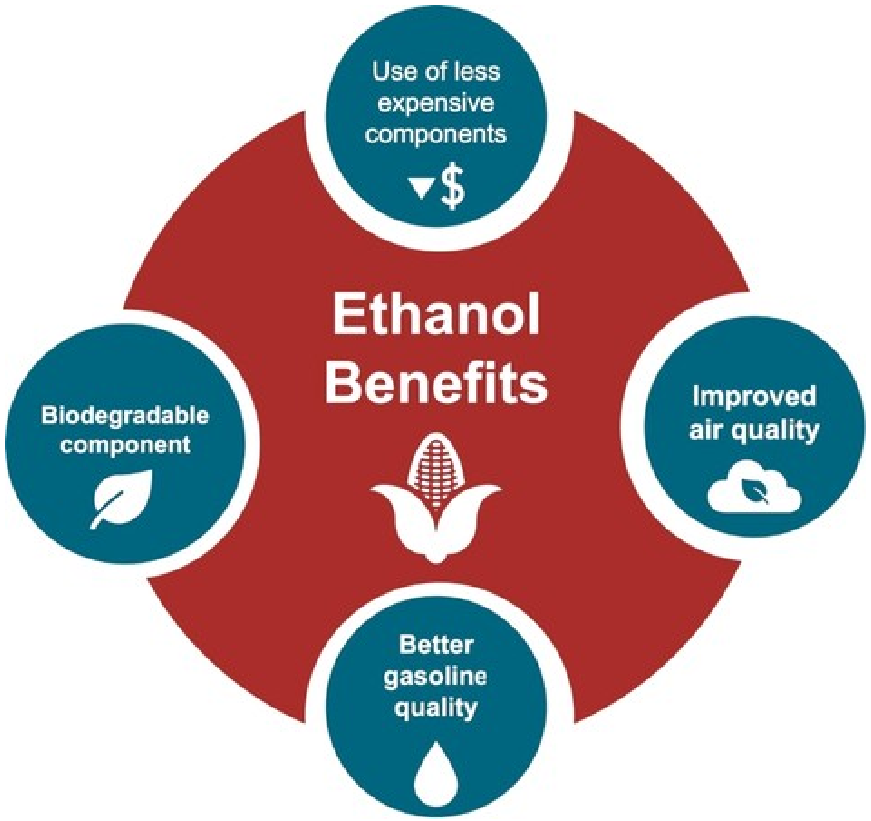
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component.The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives.To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

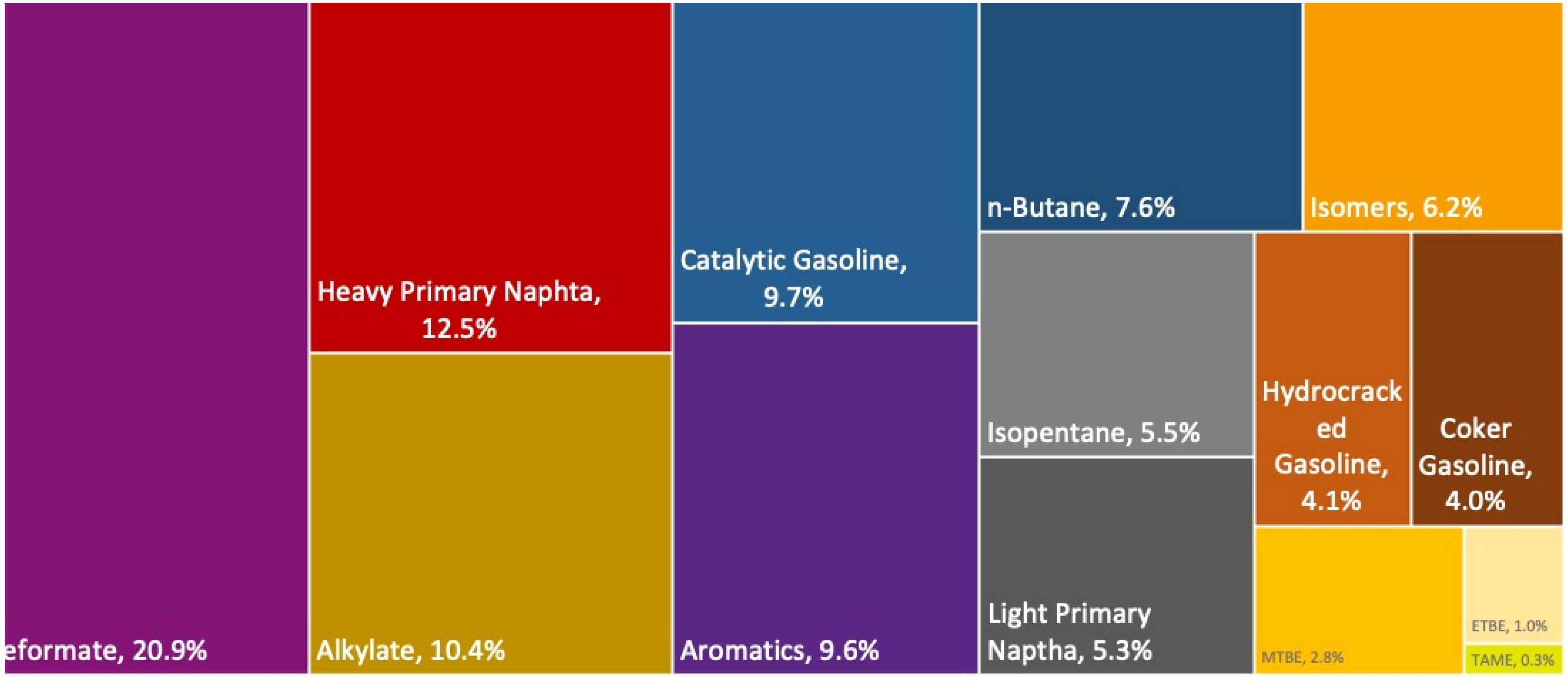
- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2025 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.

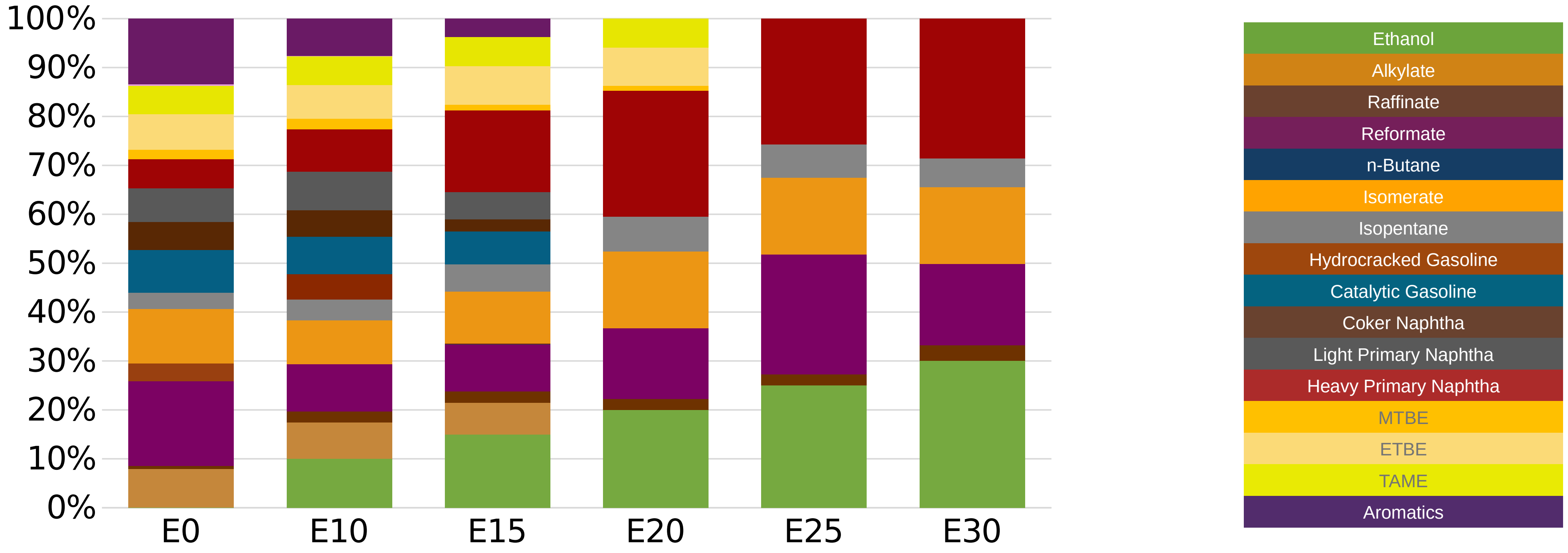


Bulgaria Available Blending Components



Source: Faro90

Bulgaria – Gasoline 95 – Constant Octane

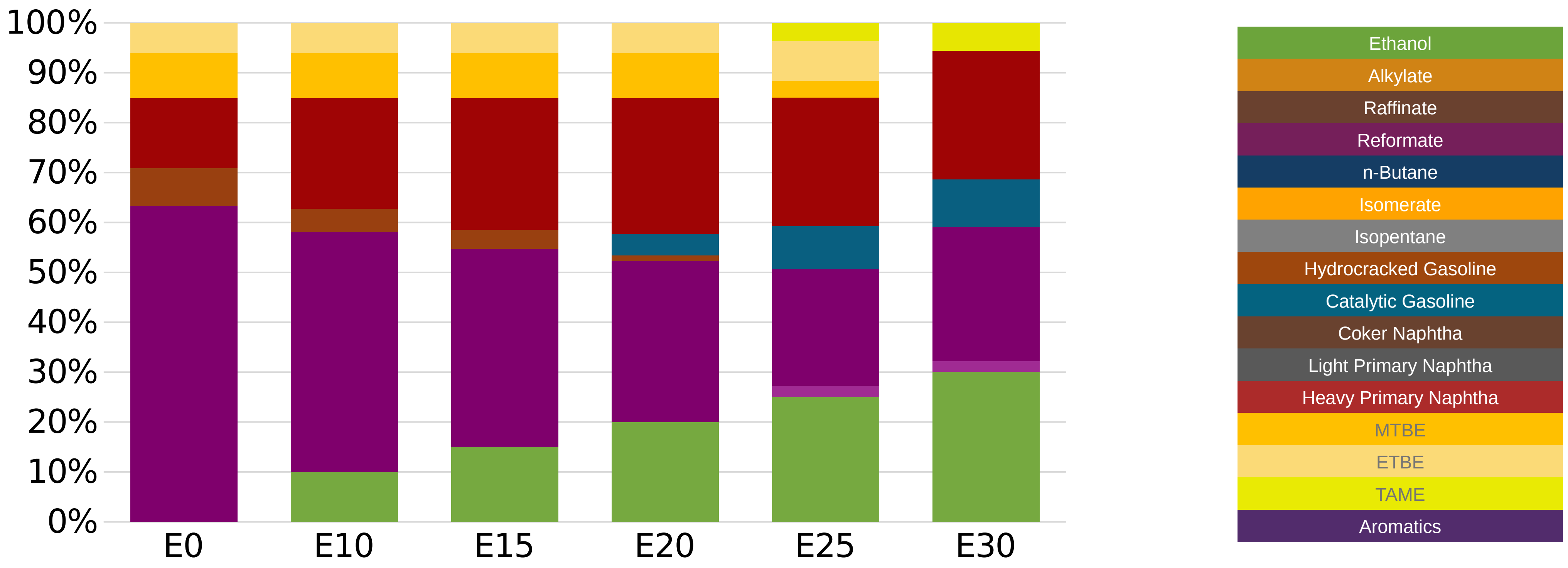


Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US$/gal)	$ 2.091	$ 1.953	$ 1.853	$ 1.738	$ 1.692	$ 1.639

Average CIF 2025 Prices

Source: Faro90

Bulgaria - Gasoline 98 - Constant Octane

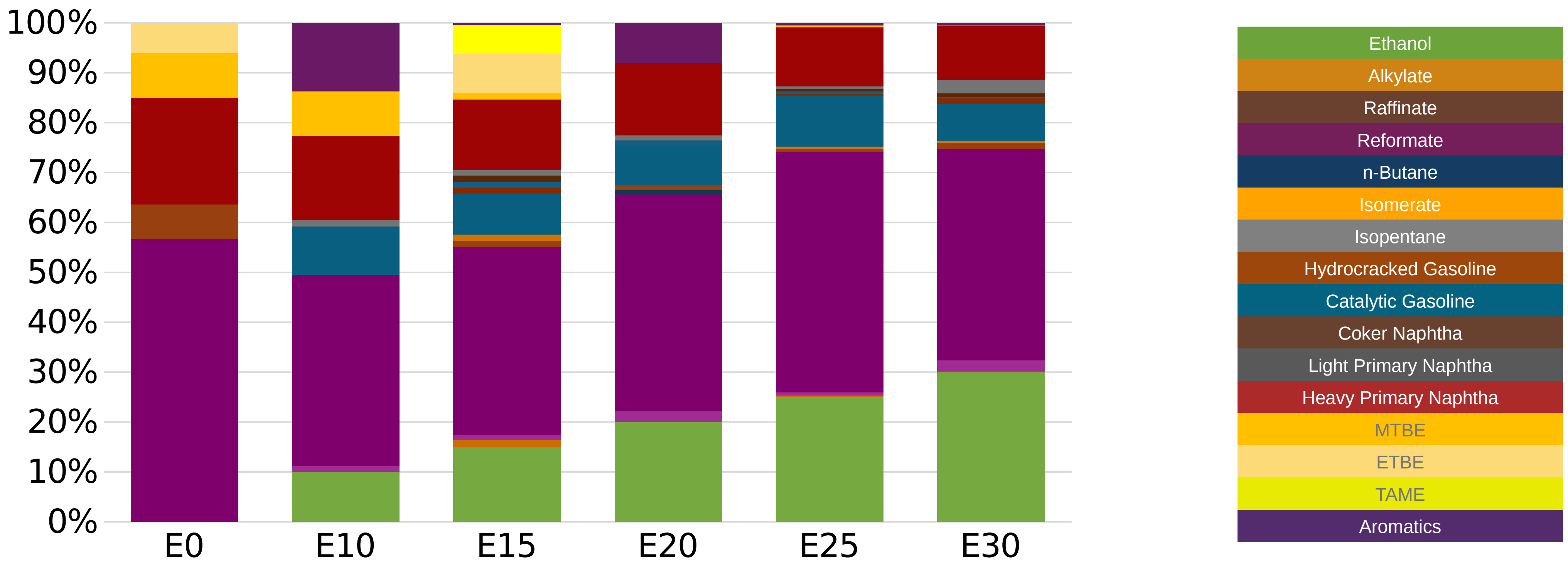


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US$/gal)	$ 2.111	$ 1.994	$ 1.925	$ 1.853	$ 1.788	$ 1.738

Average CIF 2025 Prices

Source: Faro90

Bulgaria – Gasoline 95 – Increased Octane

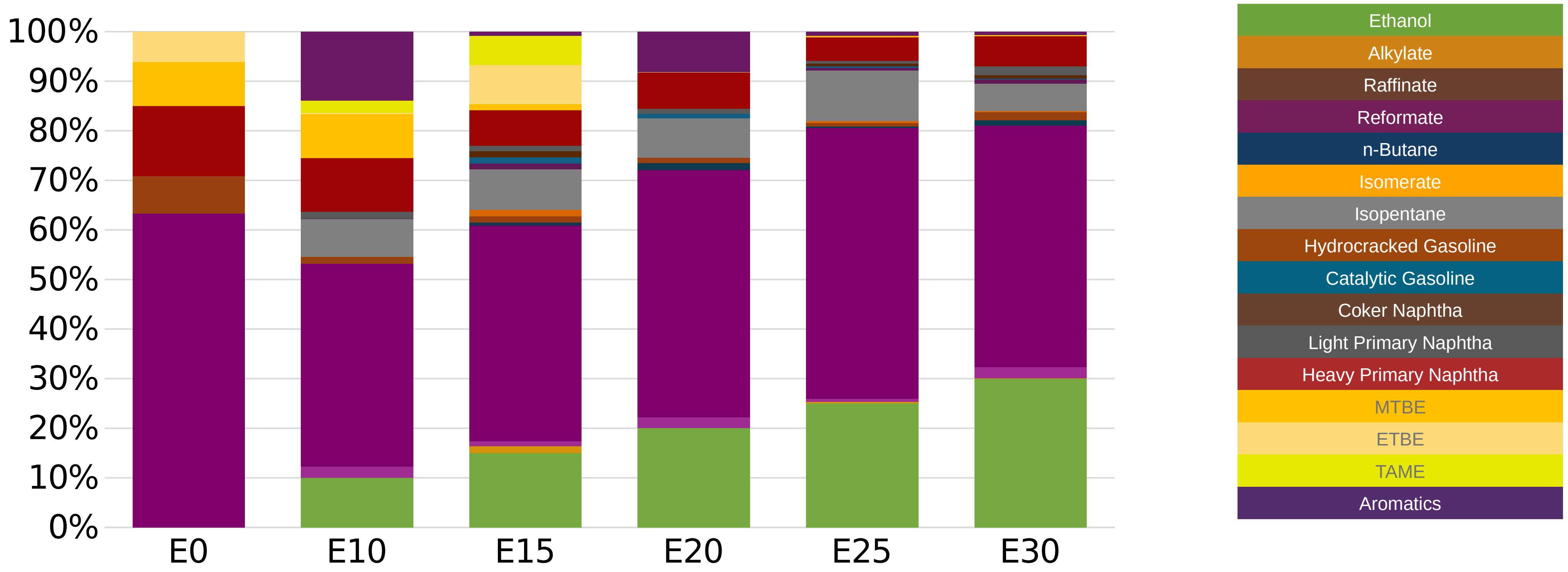


Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US$/gal)	$ 1.979	$ 1.979	$ 1.979	$ 1.979	$ 1.979	$ 1.979

Average CIF 2025 Prices

Source: Faro90

Bulgaria – Gasoline 98 – Increased Octane



Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US$/gal)	$ 2.111	$ 2.111	$ 2.111	$ 2.111	$ 2.111	$ 2.111

Average CIF 2025 Prices

Source: Faro90

Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the **International Vehicle Emissions Model (IVE).**
The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

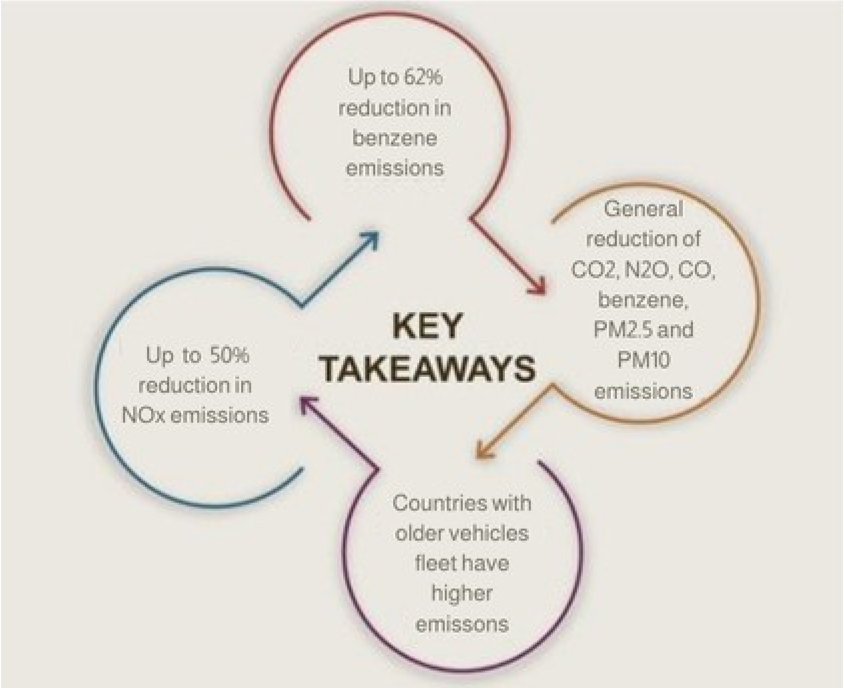
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



Gasoline Vehicle Fleet - Bulgaria

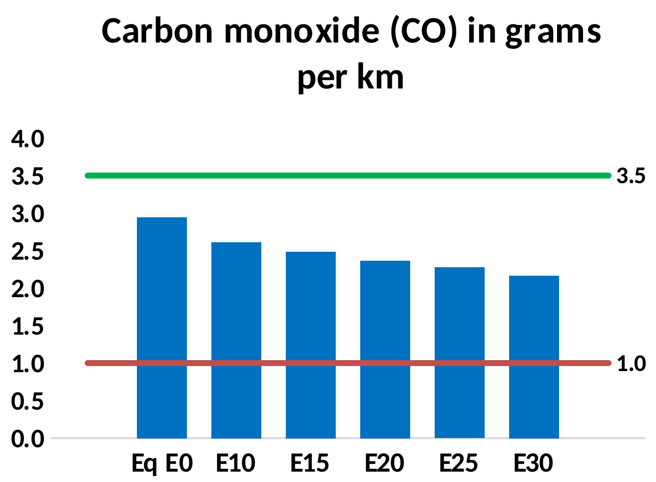
Type	Weight	Engine	Exhausto	Recovery	Age	No. Vehicles
Auto/Sml Truck	Light	Carburator	3-Ways	PCV Tank	>8 years	1,007
		Single FI	3-Ways		>8 years	22,374
		Multi FI	3-Ways		>8 years	313,475
			Euro III		>8 years	480,702
			Euro IV		>8 years	416,156
			Euro V		>8 years	278,491
			Euro 6		>8 years	546,919
					4-8 years	120,469
					<4 years	313,997
			Hybrid		All	47,300
		Electric			All	18,947
	Medium	Carburator	3-Ways	PCV Tank	>8 years	100
		Single FI	3-Ways		>8 years	646
		Multi FI	3-Ways		>8 years	37,570
			Euro III		>8 years	91,667
			Euro IV		>8 years	150,082
			Euro V		>8 years	85,708
			Euro 6		>8 years	182,306
					4-8 years	68,254
					<4 years	261,305
			Hybrid		All	17,508
		Electric			All	7,013
Sml Engines	Light	4-Cycle	None	PCV	>8 years	9,146
			Euro III		>8 years	13,242
			Euro IV		>8 years	56,417
			Euro V		>8 years	36,450
			Euro 6		>8 years	56,248
					4-8 years	64,303
					<4 years	16,122
			Hybrid		All	109
		Electric			All	2,905
Truck	Heavy	Carburator	3-Ways	PCV	>8 years	17,711
		Fuel-Injection	3-Ways		>8 years	200,161
			Euro III		>8 years	112,258
			Euro IV		>8 years	129,403
			Euro V		>8 years	36,225
			Euro 6		>8 years	51,585
					4-8 years	53,844
					<4 years	32,027
			Hybrid		All	11,507
		Electric			All	3,452

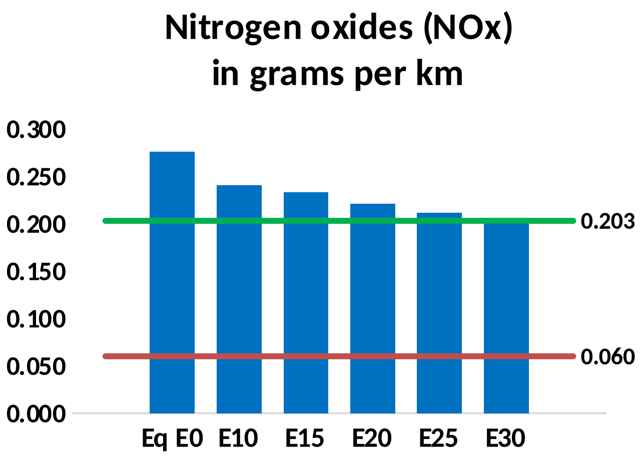
Vehicle Fleet: 4,365,110
Gasoline Vehicle Fleet: 2,909,802

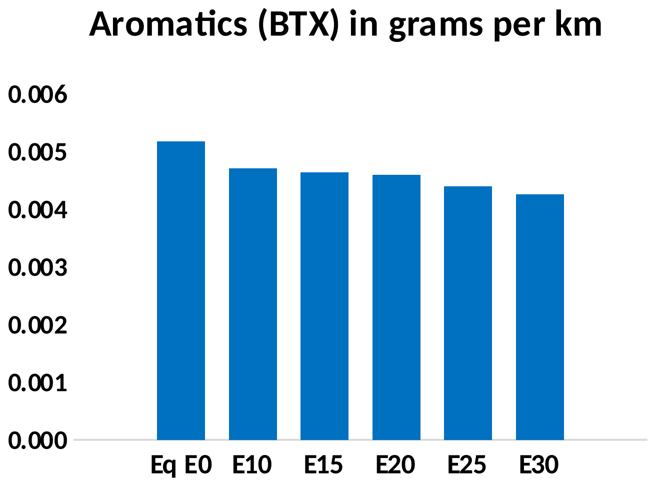
Source: Eurostat, ACEA

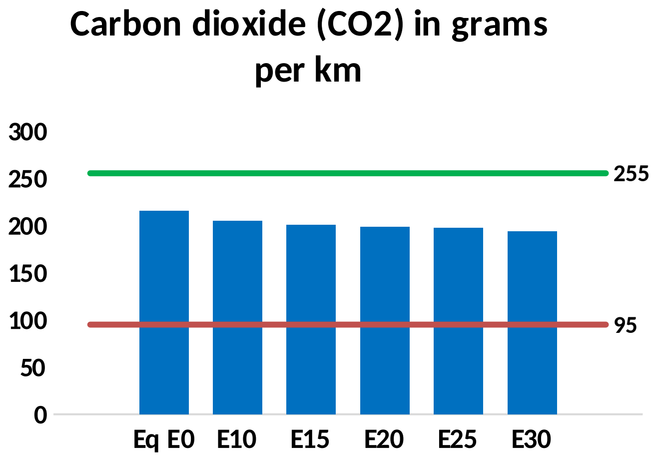
Bulgaria – Vehicular Emissions

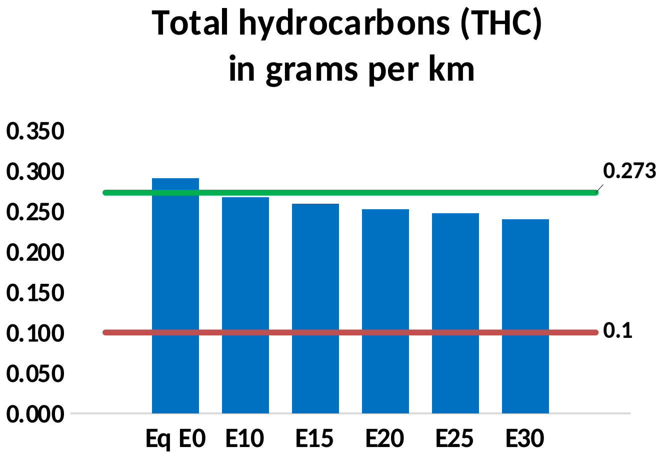
TIER III BIN 125 United States
Euro 6 Standard

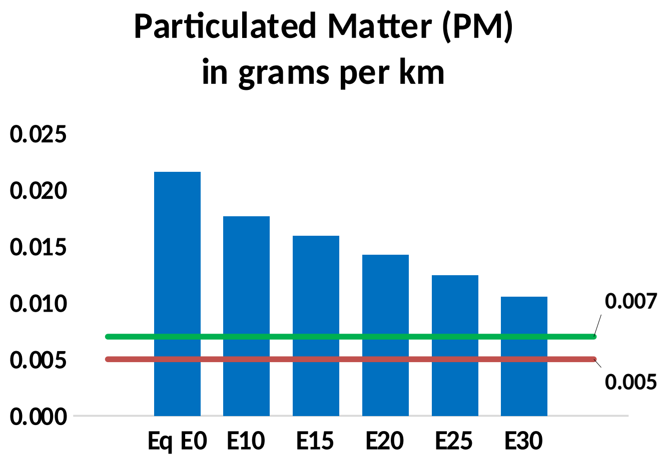












Source: Faro90

Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	218,139	205,812	193,485	184,251	175,357	168,639	160,529	-11%	-20%
CO2	15,979,135	15,568,243	15,180,179	14,874,977	14,723,951	14,636,070	14,366,285	-5%	-8%
VOC	21,538	20,654	19,771	19,197	18,689	18,319	17,752	-8%	-13%
NOx	20,472	18,902	17,825	17,277	16,393	15,700	14,884	-13%	-20%
SOx	182	169	155	143	132	120	107	-15%	-28%
NH3	5,390	5,337	5,283	5,309	5,368	5,410	5,445	-2%	0%
N2O	283	281	280	281	287	291	294	-1%	2%
CH4	4,892	4,892	4,892	4,966	5,073	5,166	5,254	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	223	218	213	211	210	210	209	-5%	-6%
Acetaldehyde	464	533	782	1,190	1,622	1,853	2,189	68%	249%
Formaldehyde	1,435	1,395	1,627	1,910	2,001	2,214	2,410	13%	39%
Benzene	383	368	349	343	341	326	315	-9%	-11%
PM2.5	926	842	758	684	611	534	452	-18%	-34%
PM10	672	611	550	496	444	387	328	-18%	-34%

Source: Faro90

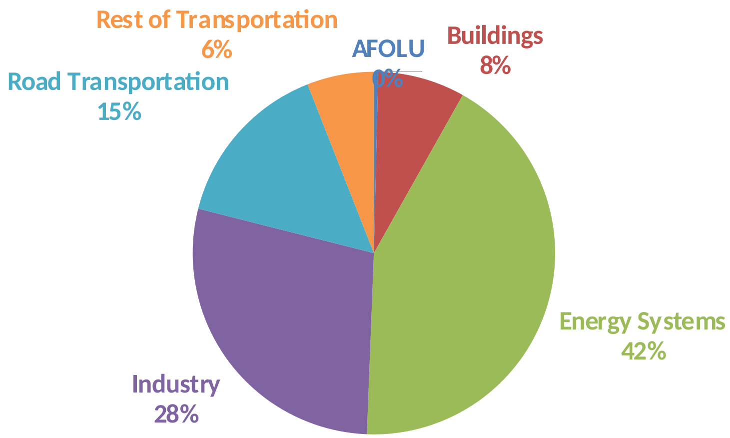
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

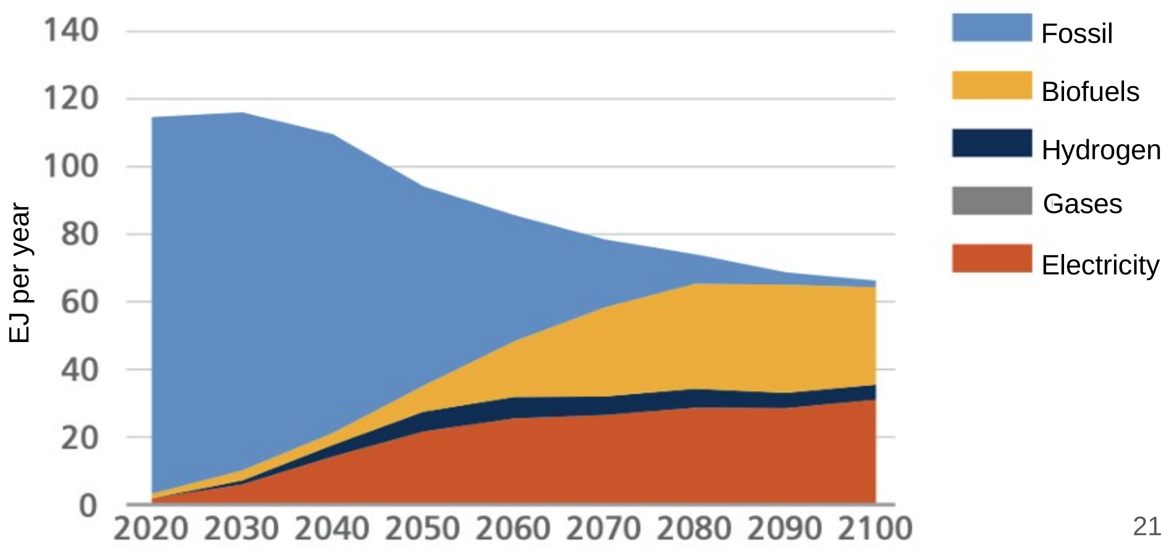
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE).** Results are compared to the current country targets for 2035. Current 2025 country emission are those published by **IPCC**.

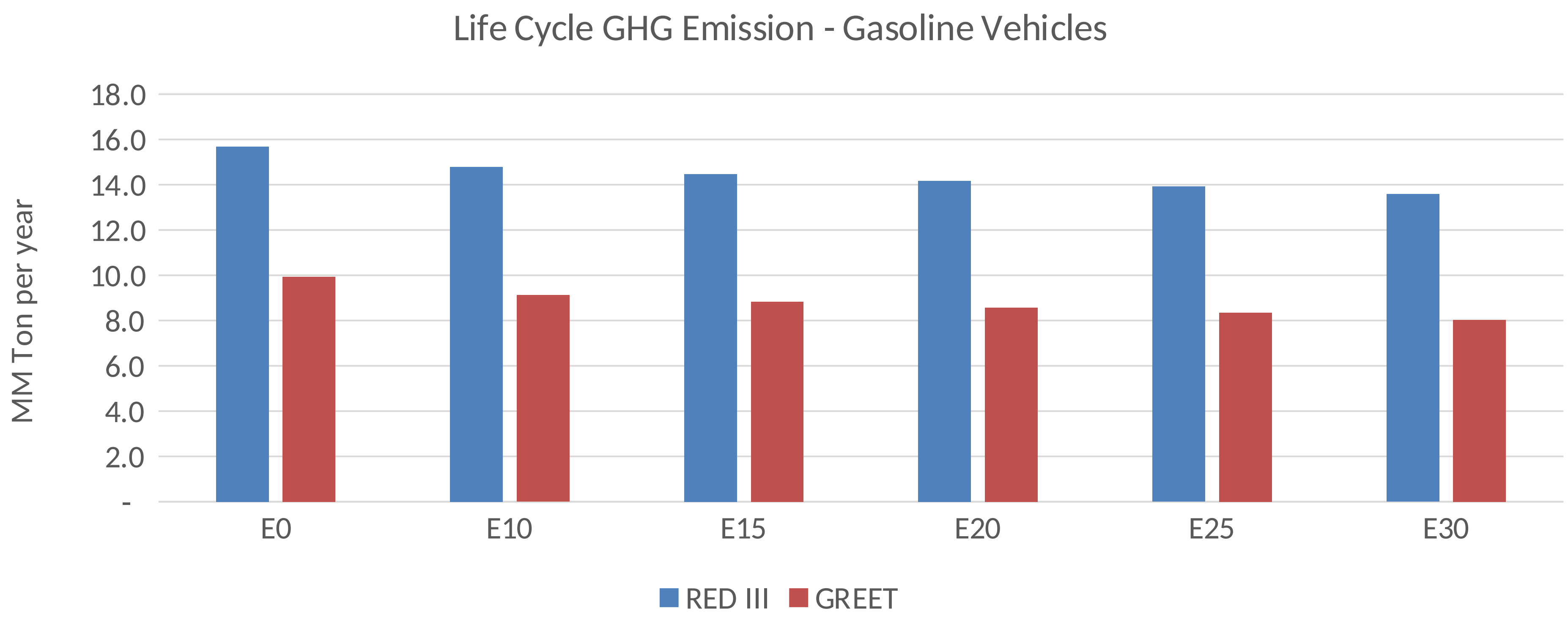
Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023



Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	15.3
Target @ 2035 (MMT/year)	3.1
Delta	12.1

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90